

Implied Vol Event Time Series

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The implied volatility should be the market's expectation of the future volatility of the underlying spot process, at the same time, current spot movement can provide information about future volatility. To model the causal structure of the underlying spot process and the implied volatility, we can consider three time series,

$$r_i = \mu_i + \tilde{\sigma}_i \epsilon_i, \quad (1)$$

$$\begin{aligned} \tilde{\sigma}_i^2 &= \tilde{\theta}_i + \sum_{j=1}^n \left(\tilde{\alpha}_j r_{i-j}^2 + \tilde{\beta}_j \tilde{\sigma}_{i-j}^2 + \tilde{\gamma}_j \sigma_{i|i-j}^2 \right) + \tilde{\nu}_i \eta_i, \\ &= \tilde{F}(r, \tilde{\phi}, \sigma), \end{aligned} \quad (2)$$

$$\begin{aligned} \sigma_i^2 &= \phi_i + \sum_{j=1}^m \left(\alpha_j r_{i-j}^2 + \beta_j \sigma_{i-j}^2 + \gamma_j \tilde{\sigma}_{i|i-j}^2 \right) + \nu_i \zeta_i, \\ &= F(r, \tilde{\sigma}, \sigma), \end{aligned} \quad (3)$$

$$(4)$$

where r_i is the spot log-return, $\tilde{\sigma}_i$ is the stepwise realized volatility, σ_i is the implied volatility with horizon at t_i and expiry at $t_{i+1} = t_i + \Delta t$ and $\sigma_{i|i-j}$ is the implied volatility with horizon at t_i and expiry at $t_i + \Delta t$ conditional on information up to t_{i-j} which for simplicity can be model by,

$$\sigma_{i|i-j}^2 = \frac{1}{\Delta t} \left[\sigma(t_{i-j}, t_{i+1})^2 (t_{i+1} - t_{i-j}) - \sigma(t_{i-j}, t_i)^2 (t_i - t_{i-j}) \right]. \quad (5)$$

The parameters $\mu_i, \tilde{\phi}_i, \phi_i, \tilde{\nu}_i, \nu_i$ can be time-varying or just constant. The parameters $\tilde{\alpha}_j, \tilde{\beta}_j, \tilde{\gamma}_j, \alpha_j, \beta_j, \gamma_j$ are constant and can be estimated using maximum likelihood estimation or Bayesian inference. The noise terms $\epsilon_i, \eta_i, \zeta_i$ are sampled from $\mathcal{N}(0, \Delta t)$.

We can include event weights in realised and implied vol by introducing a multiplicative factors, $\tilde{\omega}_i$ and ω_i , to the volatility processes,

$$\tilde{\sigma}_i^2 = \tilde{\omega}_i \tilde{F}(r, \tilde{\sigma}, \sigma), \quad (6)$$

$$\sigma_i^2 = \omega_i F(r, \tilde{\sigma}, \sigma). \quad (7)$$

one can also extend the model to include poisson jump process to the spot process like,

$$r_i = \mu_i + \tilde{\sigma}_i \epsilon_i + J_i, \quad (8)$$

$$J_i = \sum_{k=1}^{N_i} Y_k, \quad (9)$$

$$N_i \sim \text{Poisson}(\lambda \Delta t), \quad (10)$$

$$Y_k \sim \mathcal{N}(\mu_J, \sigma_J^2). \quad (11)$$

Exercise:

1. *Use Granger causality/TDMI to construct the causal structure and modify the time series model accordingly.*
2. *Calibrate the models to the implied volatility and spot log-return time series data. Measure the forecasting accuracy of the model using out-of-sample data.*
3. *Estimate the event weights using this model by first constructing the "eventless" time series by removing the targeted events, then calibrating the model to the "eventless" time series and finally using the calibrated model to estimate the event weights $\tilde{\omega}_i$ and ω_i .*
4. *Repeat the above steps for non-parametric models, including transformer that should discover the causal structure automatically.*
5. *Write the above steps into a simple function/api that can be used to estimate the event weights for any target events.*
6. *Discover hidden events in realised vol that are not captured in implied vol, vice and versa.*
7. *Highlight any new information obtained from this causal inference and event weight estimation/discovery. (We will use these to generate new trading signals.)*